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Summary

AI Engineer building production LLM systems — RAG and vector retrieval, multi-model orchestration, and content-safety guardrails — shipped as complete products rather than prototypes, with the payments, real-time dashboards, and multi-tenant APIs around them. Took six platforms to production in a single year across children's education, legal technology, generative art, AI video generation, AI knowledge management, and AI-cost attribution, including safety guardrails filtering every model response for an under-13 audience in 20+ languages. Earlier background in computer vision (YOLOv8, ANPR, OCR) deployed with a government surveillance partner.

Experience

BIG Immersive **Aug 2025 – Present**

Associate GenAI Engineer *Nov 2025 – Present*

- Ship production GenAI features end to end across **six client and internal platforms** — children's education (Yoluno), legal technology (Graysen), generative art (SoundBYTE), AI video generation (MotionForge), AI knowledge management (myNeutron), and AI-cost attribution (LLM Meter) — owning work from model integration through deployment.
- Build **RAG pipelines and vector search** over production knowledge bases: chunking and embedding of mixed-format sources, semantic retrieval over per-user corpora, and citation-grounded legal retrieval over UAE federal law.
- Orchestrate **multiple LLM and generative-media providers** — OpenAI, Gemini via OpenRouter, Replicate, and video models (Kling, Hailuo, Runway) — behind a single service layer spanning chat, vision, transcription, image generation, video generation, and TTS, so swapping a model is a config change and a hanging provider cannot pin an API worker.
- Design and implement **LLM content-safety and moderation layers**, including a multi-stage filter that screens every message inbound and every model response outbound before it reaches a child user, with a final deterministic layer that still runs when every AI service is down.
- Deliver the surrounding product surface: **multi-tenant APIs**, Stripe billing, real-time dashboards over Socket.io, and Dockerised services deployed on Railway.

AI Excellence Program *Aug 2025 – Oct 2025*

- Built a **RAG-based chatbot** delivering context-aware responses grounded in a custom document corpus.
- Contributed to AI/ML projects across model building, evaluation, and deployment, and collaborated with the team on live client work.

FYP Collaboration — Punjab Safe Cities Authority (PSCA) Nov 2024 – Jan 2025

AI Engineer (Final Year Project) *Lahore, Pakistan*

- Built a **multi-stage computer vision pipeline** for automated litter-offence detection: **YOLOv8s** for littering-action detection, **YOLOv8n** for offending-vehicle tracking, then **ANPR** and **OCR** to extract plate numbers and generate challans automatically.
- Scraped and curated a custom **30,000+ image dataset**, fine-tuned both YOLO models, and validated real-time performance against live road footage with PSCA's operations team.

Technical Skills

GenAI & LLM: RAG, vector search, embeddings, semantic memory, chunking strategy, prompt engineering, multi-model orchestration (OpenRouter), content moderation & safety guardrails, video generation (Kling, Hailuo, Runway), TTS, LangChain, OpenAI, Gemini, Anthropic, Replicate

ML & Computer Vision: PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, OpenCV, YOLOv8, ANPR, OCR, dataset curation & fine-tuning

Languages: Python, TypeScript, JavaScript, Java, C/C++, SQL, HTML/CSS

Backend: FastAPI, Node.js, Express, Next.js, REST APIs, Prisma, Socket.io, JWT/OAuth, Stripe, Docker

Frontend: React, Next.js, SvelteKit, Vite, Tailwind CSS, Radix/shadcn, TanStack Query, Zustand

Data & Infrastructure: PostgreSQL, MongoDB, MySQL, Redis, Firebase, AWS S3, Railway, Vercel

Practices: Multi-tenant architecture, rate limiting, webhooks, E2E testing (Playwright), Git

Projects

Yoluno | *React, TypeScript, Express, PostgreSQL, Gemini via OpenRouter, Socket.io, Stripe* yoluno.com

- Child-safe AI learning platform for ages 3–13: age-adaptive AI companions, illustrated story generation, and family-history features across 20+ languages, grounded in the English National Curriculum and IB (KS1–KS4).
- Built a **three-layer content-safety pipeline** — banned-topic matching, keyword expansion, and typo-tolerant fuzzy matching — classifying every interaction green/yellow/red and escalating flagged content to parents in real time over Socket.io. Fuzzy matching needed three guards (stop-word blocklist, length-similarity bound, prefix rule) to stop "can you teach me" matching the blocked topic "cats".
- Orchestrated **multiple Gemini models via OpenRouter** for chat, vision, story text, illustration, and transcription, with OpenAI TTS for narration and per-call abort timeouts protecting the API workers.

MotionForge (Stytch Me Up) | *Next.js, TypeScript, multi-model orchestration, Kling/Hailuo/Runway, TTS, Railway*live demo

- **Brief-to-video generation platform**: turns a text brief into a finished, on-brand marketing video — story, script, motion, voiceover, and captions generated and assembled automatically, rendered to 4K in ∼**90 seconds**.
- Orchestrated **multi-provider video generation behind one gateway** — Kling, Hailuo, and Runway — with image-to-video and end-frame anchoring, so choosing a model is a config change and a slow provider degrades one shot rather than the whole render.
- Enforced **brand-locked output** (colours, fonts, tone) as a scored constraint downstream of generation, with word-level karaoke caption sync driven by TTS timestamps and rendering run as an async job producing MP4/WebM/4K.

Graysen | *RAG, vector retrieval, citation grounding, bilingual NLP* graysen.ai

- AI contract analyser for UAE law that flags problematic clauses — hidden renewals, excessive penalties, ambiguous terms — and returns a contract health score, with AI-suggested rewrites as negotiation starting points.
- Built **clause-level retrieval grounded in UAE federal law with source citations**: responses are constrained to what retrieval actually returned, so no claim ships without the source it came from. Bilingual English/Arabic handling sits in the retrieval path rather than bolted on as translation.

myNeutron | *RAG, vector search, semantic memory, embeddings, Chrome extension* myneutron.ai

- Persistent semantic memory shared across ChatGPT, Gemini, and email — context follows the user rather than living inside whichever assistant they happened to use.
- Built the **semantic memory and retrieval modules**: chunking and embedding of mixed-format sources, vector search over each user's corpus, and rank-and-trim retrieval that treats the context window as a budget rather than storing everything and hoping.

SoundBYTE© / Sonic Identity™ | *React monorepo, Express, Prisma, Replicate, Gemini, Stripe, Redis* amritasethi.com

- Voice-to-art generative platform for a Dubai artist featured on CNN, Forbes, and BBC: captures a voice recording in-browser via MediaRecorder and transforms it into a unique generative visual identity.
- Shipped a **3-app npm-workspaces monorepo** (marketing site, creation playground, admin console) over a shared Express + Prisma API with Stripe payments, Redis sessions, S3 storage, and print-on-demand fulfilment — the generated image is not the deliverable, a framed print arriving at a door is.

LLM Meter | *Next.js 15, Prisma, PostgreSQL, Recharts, multi-tenant* *internal*

- Cross-provider AI spend **attribution** platform answering which team, project, or person spent what on AI — tracking OpenAI, Anthropic, Gemini, OpenRouter, fal.ai, and GLM in one place.
- Built API-key-authenticated multi-tenant ingestion with daily rollups, reconciling three overlapping capture paths that see the same tokens and must never be summed, plus a **subscription seat-cost allocation model** that attributes flat-fee AI seats down to per-project cost.

Sales Pitch Generator | *SvelteKit, FastAPI, Gemini API, RAG*

- AI platform that automatically generates sales pitch presentations, using RAG over company material and Gemini for contextual content generation.

Education

FAST NUCES, Lahore Aug 2021 – July 2025
Bachelor of Science in Computer Science *Lahore, Pakistan*

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Computer Vision, Data Science, Data Structures & Algorithms, Database Systems, Operating Systems, Computer Networks.

Achievements

- Shipped **six production AI platforms** within the first year as an engineer — spanning children's education, legal technology, generative art, AI video generation, AI knowledge management, and internal cost tooling.
- Designed and shipped **LLM safety guardrails for a children's product**, classifying every model interaction across three detection layers before it reaches a user aged 3–13, in 20+ languages.
- Built a **subscription seat-cost attribution model** that allocates flat-fee AI seats down to per-project cost — a capability not available in any off-the-shelf AI spend tool.
- Curated a **30,000+ image training dataset** and fine-tuned YOLOv8 detection and tracking models, achieving real-time performance on live road footage for a government surveillance deployment.